

Exercises (Involving Residues over a Prime)

1. Let p be an odd prime, $1 - \frac{1}{2^2} + \frac{1}{3^2} - \dots + (-1)^{p-2} \frac{1}{(p-1)^2} \equiv 0 \pmod{p}$.
2. Let p be an odd prime and $p \geq 5$. Suppose $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} - \dots + \frac{1}{(p-1)^2} = \frac{a}{b}$, then $p \mid a$.
3. (Wolstenholme) If p is a prime greater than 3, then the numerators of the fraction

$$1 + \frac{1}{2} + \dots + \frac{1}{p-1}$$

is divisible by p^2 .

4. (Hardy and Wright p.104) If p is an odd prime, then

$$\frac{2^{p-1} - 1}{p} = 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{p-2} \pmod{p}.$$

5. (IMO1995.6) Let p be an odd prime number. Find the number of subsets A of the set $\{1, 2, \dots, 2p\}$ such that
 - (a) A has exactly p elements, and
 - (b) the sum of all the elements in A is divisible by p .

6. Let p be an odd prime. Show that

$$1^{p-2} + 2^{p-2} + 3^{p-2} + \dots + \left(\frac{p-1}{2}\right)^{p-2} \equiv \frac{2-2^p}{p} \pmod{p}.$$

7. Let p be an odd prime. Show that $\sum_{k=1}^{p-1} (k-1)!(p-k)!k^{p-1} \equiv 0 \pmod{p}$.
8. Let $p \geq 5$ be a prime, show that if $1 + \frac{1}{2} + \dots + \frac{1}{p} = \frac{a}{b}$, then $p^4 \mid ap - b$.
9. (IOM3: Problem 3) Let k be a positive integer such that $p = 8k + 5$ is a prime number. The integers $r_1, r_2, \dots, r_{2k+1}$ are chosen so that the numbers $0, r_1^4, r_2^4, \dots, r_{2k+1}^4$ give pairwise different remainders modulo p . Prove that

$$\prod_{1 \leq i, j \leq 2k+1} (r_i^4 + r_j^4) \equiv (-1)^{k(k+1)/2} \pmod{p}.$$

10. (IOM4: Problem 6) Let p be a prime number and let $f(x)$ be a polynomial of degree d with integer coefficients. Assume that the numbers $f(1), f(2), \dots, f(p)$ leave exactly k distinct remainders when divided by p , and $1 < k < p$. Prove that

$$\frac{p-1}{d} \leq k-1 \leq (p-1)\left(1 - \frac{1}{d}\right).$$

11. Let p be an odd prime and set $\omega = \exp(2\pi i / p)$. Evaluate the product

$$E(p) = (\omega^{r_1} + \omega^{r_2} + \dots + \omega^{r_{(p-1)/2}})(\omega^{n_1} + \omega^{n_2} + \dots + \omega^{n_{(p-1)/2}}),$$

where $r_1, r_2, \dots, r_{(p-1)/2}$ denote the $(p-1)/2$ quadratic residues modulo p and

$n_1, n_2, \dots, n_{(p-1)/2}$ denote the $(p-1)/2$ quadratic nonresidues modulo p

February 2020